

Central-Potential Scattering, Rutherford Cross Section, and Hamiltonian Mechanics

HC 6001 — Classical Mechanics

October 22, 2024

1 Scattering by a central potential — setup

Definition 1 (Scattering by a central potential). *Consider scattering by a central potential $V(r)$. The incident energy and velocity are*

$$E = \frac{1}{2}mv_0^2, \quad v_0 = \sqrt{\frac{2E}{m}}. \quad (1)$$

The orbital angular momentum and impact parameter are related by

$$l = msv_0 = s\sqrt{2mE}, \quad s = \frac{l}{\sqrt{2mE}}. \quad (2)$$

The volume element in spherical coordinates is $d^3r = r^2 dr d\Omega$, with $d\Omega = \sin\theta d\theta d\phi$. By azimuthal symmetry, the relevant solid-angle element for scattering into angle θ is

$$d\Omega = 2\pi \sin\theta d\theta. \quad (3)$$

2 Differential and total cross section

Let I denote the incident intensity (number of particles incident per unit time per unit area). The differential cross section $\sigma(\Omega)$ is defined by

$$\sigma(\Omega) d\Omega = \frac{\# \text{ particles scattered into } d\Omega \text{ per unit time}}{I}. \quad (4)$$

Particles incident in a ring between impact parameters s and $s + ds$ have area $2\pi s |ds|$. Flux conservation gives

$$I(2\pi s)|ds| = I\sigma(\theta)2\pi \sin\theta |d\theta|, \quad (5)$$

hence

$$\sigma(\theta) = \frac{s}{\sin\theta} \left| \frac{ds}{d\theta} \right|. \quad (6)$$

The total cross section is

$$\sigma_T = \int d\Omega \sigma(\Omega). \quad (7)$$

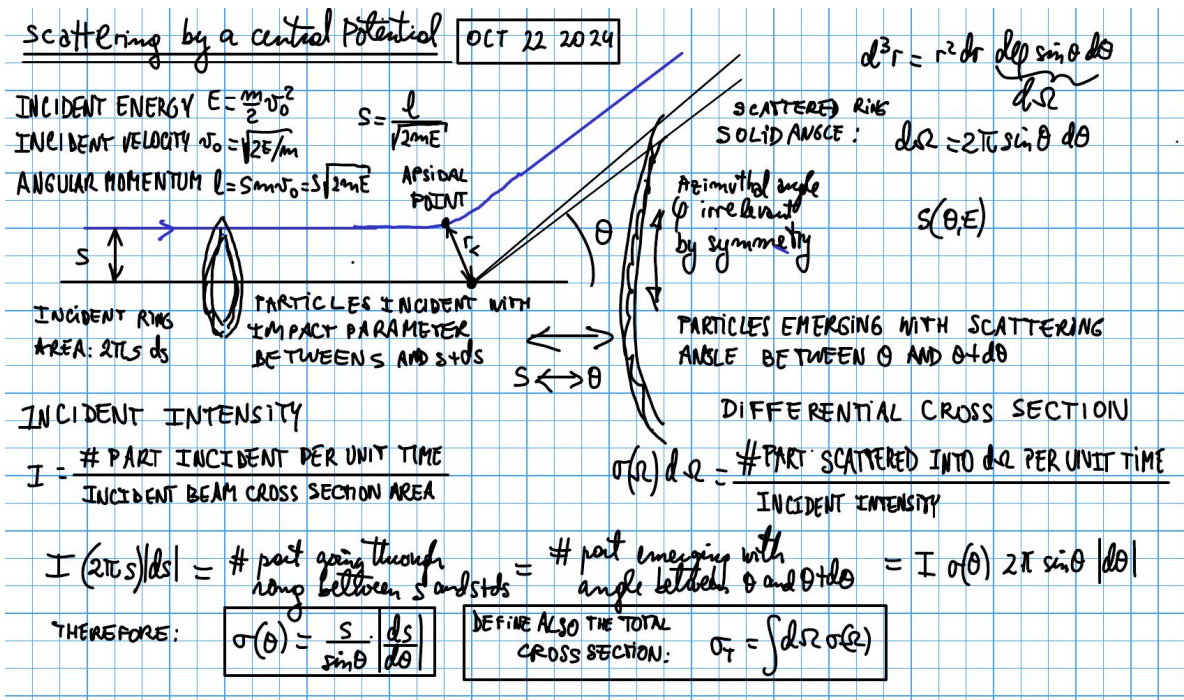


Figure 1: Scattering trajectory: impact parameter s , apsidal point at r_c , and scattering angle θ .

COMPUTING $S(\theta, E)$ (UNIVERSAL METHOD)

$E = \frac{m}{2} \dot{r}^2 + V(r) + \frac{L^2}{2mr^2}$ $\dot{r} = \left[\frac{2}{m} \left(E - V(r) - \frac{L^2}{2mr^2} \right) \right]^{1/2}$ $\dot{r} = \frac{dr}{dt} \Rightarrow dt = \frac{dr}{\dot{r}}$

$L = mr^2 \dot{\theta} \Rightarrow \frac{d\theta}{dt} = \dot{\theta} = \frac{L}{mr^2} \Rightarrow d\theta = \dot{\theta} dt = \frac{L}{mr^2} dt = \frac{L}{mr^2} \frac{dr}{\dot{r}} = \frac{L}{mr^2} \frac{dr}{\left[\frac{2}{m} \left(E - V(r) - \frac{L^2}{2mr^2} \right) \right]^{1/2}} = \frac{L}{\sqrt{2E}} \frac{r^2 dr}{\left[1 - \frac{V(r)}{E} - \frac{L^2}{2mE r^2} \right]^{1/2}}$

Figure 2: Repulsive and attractive central-potential trajectories. Let ψ be the angle from the horizontal to the apsidal direction. Repulsive: $\theta = \pi - 2\psi$; attractive: $\theta = 2\psi - \pi$; in all cases $\theta = |\pi - 2\psi|$.

3 Repulsive vs attractive scattering geometry

4 Universal scattering-angle integral for any $V(r)$

To compute $S(\theta, E)$ (equivalently $s(\theta, E)$) for any central potential, start from

$$E = \frac{1}{2}m\dot{r}^2 + V(r) + \frac{l^2}{2mr^2}. \quad (8)$$

Solving for the radial velocity,

$$\dot{r} = \sqrt{\frac{2}{m} \left(E - V(r) - \frac{l^2}{2mr^2} \right)}, \quad dt = \frac{dr}{\dot{r}}. \quad (9)$$

With $l = mr^2\dot{\theta}$,

$$d\theta = \dot{\theta} dt = \frac{l}{mr^2} dt = \frac{l r^{-2} dr}{\sqrt{\frac{2m}{E} \left(1 - \frac{V(r)}{E} - \frac{l^2}{2mEr^2} \right)}}. \quad (10)$$

Using $l = s\sqrt{2mE}$,

$$d\theta = \frac{s r^{-2} dr}{\sqrt{1 - \frac{V(r)}{E} - \frac{s^2}{r^2}}}. \quad (11)$$

The apsidal angle is

$$\psi = \int_{r_m}^{\infty} \frac{s r^{-2} dr}{\sqrt{1 - \frac{V(r)}{E} - \frac{s^2}{r^2}}}. \quad (12)$$

With $u = 1/r$ and $du = -r^{-2} dr$,

$$\psi = \int_0^{u_m} \frac{s du}{\sqrt{1 - \frac{V(u)}{E} - s^2 u^2}}, \quad (13)$$

where $u_m = 1/r_m$.

$$\theta(s) = \left| \pi - 2 \int_0^{u_m} \frac{s du}{\sqrt{1 - \frac{V(u)}{E} - s^2 u^2}} \right|. \quad (14)$$

From $\theta(s)$ one obtains $|ds/d\theta|$ and hence $\sigma(\theta)$ via (6). **Up to here, results are valid for any central potential $V(r)$.**

5 Coulomb scattering — general hyperbolic orbits

Definition 2 (Coulomb scattering). *For Coulomb scattering,*

$$f = \frac{ZZ'e^2}{r^2}, \quad V(r) = -\frac{k}{r}, \quad k = -ZZ'e^2. \quad (15)$$

For $V = -k/r$ only:

- $E > 0$: hyperbolic orbit

- $E = 0$: *parabolic orbit*
- $E < 0$: *elliptic orbit*

For scattering ($E > 0$), the orbit equation is

$$\frac{1}{r} = C[1 + \epsilon \cos(\theta - \theta')], \quad (16)$$

with

$$C = \frac{mk}{l^2}, \quad \epsilon = \left[1 + \frac{2El^2}{mk^2}\right]^{1/2} = \left[1 + \left(\frac{2sE}{ZZ'e^2}\right)^2\right]^{1/2} > 1, \quad (17)$$

using $l = s\sqrt{2mE}$.

6 Coulomb scattering — repulsive case

Case i: repulsive ($ZZ' > 0$, so $k < 0$ and $C < 0$). Take periapsis at $\theta' = \pi$:

$$\frac{1}{r} = \frac{mZZ'e^2}{l^2}(\epsilon \cos \theta - 1). \quad (18)$$

The apsidal point is at $\theta = 0$, with

$$\frac{1}{r_m} = \frac{mZZ'e^2}{l^2}(\epsilon - 1). \quad (19)$$

As $r \rightarrow \infty$, $\cos \psi = 1/\epsilon$. With $\psi = (\pi - \theta)/2$,

$$\frac{1}{\epsilon} = \sin \frac{\theta}{2}. \quad (20)$$

7 Coulomb scattering — attractive case

Case ii: attractive ($ZZ' < 0$, so $k > 0$; also gravity with $k = Gm_1m_2$). Take $\theta' = 0$:

$$\frac{1}{r} = C[1 + \epsilon \cos \theta]. \quad (21)$$

The apsidal point is at $\theta = 0$. As $r \rightarrow \infty$, $\cos \psi = -1/\epsilon$. With $\psi = (\pi + \theta)/2$,

$$\frac{1}{\epsilon} = \sin \frac{\theta}{2}, \quad (22)$$

the same relation as in the repulsive case.

8 Rutherford differential cross section

Combining both Coulomb cases,

$$\cot^2 \frac{\theta}{2} = \frac{1}{\sin^2 \frac{\theta}{2}} - 1 = \epsilon^2 - 1 = \left(\frac{2sE}{zz'e^2}\right)^2, \quad (23)$$

so

$$s(\theta, E) = \left| \frac{zz'e^2}{2E} \right| \cot \frac{\theta}{2}. \quad (24)$$

Then

$$\frac{ds}{d\theta} = - \left| \frac{zz'e^2}{4E} \right| \frac{1}{\sin^2 \frac{\theta}{2}}, \quad (25)$$

and using (6),

$$\sigma(\theta) = \frac{1}{4} \left(\frac{zz'e^2}{2E} \right)^2 \csc^4 \frac{\theta}{2}. \quad (26)$$

This is the **Rutherford scattering cross section** (for Coulomb force).

Remark 1. 1. *The cross section is the same for repulsive and attractive cases, since it is proportional to $(zz')^2$.*

2. *It diverges like E^{-2} at low energies.*

3. *It diverges like θ^{-4} (very strong) for small angles θ .*

4. *The classical result is identical to the quantum-mechanical result.*

The total cross section is

$$\sigma_T = \int \sigma(\Omega) d\Omega. \quad (27)$$

Substituting (26) and $d\Omega = 4\pi \sin \frac{\theta}{2} \cos \frac{\theta}{2} d\theta$,

$$\sigma_T = \pi \left(\frac{zz'e^2}{2E} \right)^2 \int_0^\pi \frac{\sin \frac{\theta}{2} \cos \frac{\theta}{2}}{\sin^4 \frac{\theta}{2}} d\theta. \quad (28)$$

For small θ , the integrand behaves like $1/\theta^3$, so the integral diverges; σ_T is **strongly divergent**.

9 Notation, divergent total cross section, screened potentials

Remark 2 (Notation and long-range potentials). *Goldstein uses $\sigma(\Omega)$ and σ_T for the differential and total cross sections; other texts use $d\sigma/d\Omega$ and σ , respectively.*

Classically, $\sigma_T = +\infty$ for any long-range potential. In quantum mechanics, $\sigma_T = +\infty$ for some long-range potentials $V(r)$ that do not decay fast enough as $r \rightarrow \infty$.

Caveat: *In a dense conducting material, the Coulomb potential becomes a screened (Yukawa) potential:*

$$V_{\text{screen}}(r) = -\frac{k}{r} e^{-r/\lambda}. \quad (29)$$

In non-conducting materials the screened potential may differ but still vanishes much faster than $1/r$.

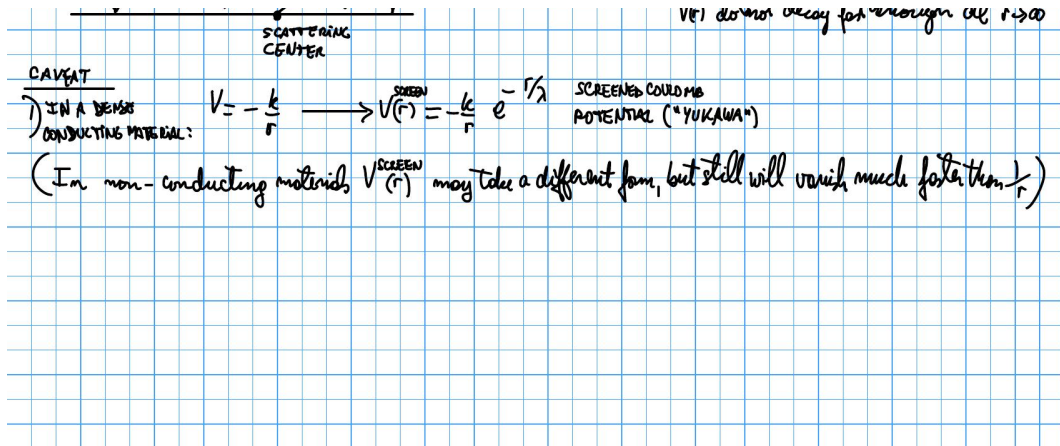


Figure 3: Scattering geometry and notation comparison (from block notes).

10 Hamiltonian mechanics — Lagrangian review

Definition 3 (Lagrangian formalism). *The Lagrangian is $L = T - V = L(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n, t)$. The Euler–Lagrange equations are*

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} = 0, \quad (30)$$

with generalized momentum $p_i = \partial L / \partial \dot{q}_i$.

Remark 3. In polar coordinates, $\partial L / \partial \dot{\theta} = p_\theta = l_z$.

11 Legendre transformation (math and thermodynamics)

We wish to transform coordinates from $(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n)$ to $(q_1, \dots, q_n, p_1, \dots, p_n)$.

For a function $f(x, y)$, $df = u dx + v dy$ with $u = \partial f / \partial x$ and $v = \partial f / \partial y$.

Thermodynamics example (fluid at fixed particle number N): the first law is $dU = \delta Q + \delta W = T dS - p dV$, with $U = U(S, V)$. To change variables to (T, V) , define the Helmholtz free energy $A = U - TS$, giving $dA = -S dT - p dV$, so $A = A(T, V)$.

General Legendre transform: to transform $(x, y) \rightarrow (u, y)$, define $g = f - ux$. Then

$$dg = df - d(ux) = -x du + v dy, \quad (31)$$

so $g = g(u, y)$.

12 Hamiltonian and canonical equations of motion

Apply the Legendre transform to the Lagrangian with $p_i = \partial L / \partial \dot{q}_i$:

$$H = \sum_i p_i \dot{q}_i - L. \quad (32)$$

Expanding dL and $d(\sum_i p_i \dot{q}_i)$,

$$dH = -\frac{\partial L}{\partial t} dt + \sum_i \left(\dot{q}_i dp_i - \frac{\partial L}{\partial q_i} dq_i \right), \quad (33)$$

so $H = H(p_1, \dots, p_n, q_1, \dots, q_n, t)$.

Theorem 1 (Canonical equations of motion).

$$\frac{\partial H}{\partial p_i} = \dot{q}_i, \quad (34)$$

$$\frac{\partial H}{\partial q_i} = -\frac{\partial L}{\partial q_i} = -\dot{p}_i \quad (\text{Lagrange equation}), \quad (35)$$

$$\frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}. \quad (36)$$

Handwritten notes on a blue grid background showing the canonical equations of motion. The equations are: $\frac{\partial H}{\partial p_i} = \dot{q}_i$, $\frac{\partial H}{\partial q_i} = -\frac{\partial L}{\partial q_i} = -\dot{p}_i$, and $\frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}$. A bracket under the first two equations is labeled "Lagrange Eqn". A bracket under all three equations is labeled "CANONICAL EQS OF MOTION".

Figure 4: Canonical equations of motion (from lecture notes).